Uber Convertible Notes — Fair Value Memo

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Purpose and conclusions

This memo re-performs the fair value of Uber Technologies' two convertible notes outstanding during 2024, using only public information. It also documents the ASC 815 classification analysis behind Uber's accounting. Fair value of the notes is a required ASC 820 disclosure (Level 2). The notes themselves are carried at amortized cost.

Conclusion. A Tsiveriotis-Fernandes binomial lattice reproduces Uber's disclosed fair values **with no input fitted to them**. Volatility is Uber's trailing 36-month realized volatility, and the credit spread is the ICE BofA index matching Uber's rating at each date. **Seven of eight** model values fall inside the ±$0.05bn rounding band of the reported figures. The mean absolute error against the band midpoints is 2.1%.

Date	2028 Notes: model	Reported	2025 Notes: model	Reported
28-Mar-2024	$2.19bn	$2.1bn ✗	$1.27bn	$1.3bn ✓
28-Jun-2024	$2.11bn	$2.1bn ✓	$1.24bn	$1.2bn ✓
30-Sep-2024	$2.18bn	$2.2bn ✓	$1.26bn	$1.3bn ✓
31-Dec-2024	$1.91bn	$1.9bn ✓	$1.17bn	$1.2bn ✓

Model values from the 1,000-step lattice. ✓ = inside the rounding band.

- **The one miss** (2028 Notes, March 2024) followed a 22% rally in February. The reported value implies about 35% volatility, against 45% trailing realized volatility.
- **Day-one evidence:**
  - The capped calls cost $5.93 per share, more than any flat volatility can produce (maximum $4.90). The price is consistent with 46% volatility plus about 3 points of skew.
  - The notes were sold at par against a model value of about 103.6, a typical new-issue discount.

- **Accounting:** the conversion option and the capped calls are indexed to Uber's stock and equity-classified under ASC 815-40. They are not bifurcated or remeasured. Had the option failed those tests, Uber would have carried a $450–840M derivative, with quarterly earnings swings of up to about $260M.
- **Deliverables:**
  - Excel model with a live 100-step lattice for each case: 142,523 formulas, 0 errors, and an exact match to Python.
  - Python validation notebook.
  - This memo.

Instruments and terms

Both notes are senior unsecured. On conversion, Uber pays principal in cash and settles any excess in cash or shares at its election (net-share settlement). Holders can put the notes only on a fundamental change. Uber pays no dividend on its common stock.

Term	2028 Notes	2025 Notes
Principal	$1,725M (incl. $225M option)	$1,150M
Coupon	0.875%, 1 Jun / 1 Dec	0%
Issued / matures	24-Nov-2023 / 1-Dec-2028	11-Dec-2020 / 15-Dec-2025
Conversion rate	13.7848 shares per $1,000	12.3701 shares per $1,000
Conversion price	$72.54 (32.5% premium to $54.75)	$80.84 (52.5% premium)
Soft call	From 5-Dec-2026 if stock ≥ 130% of conversion price ($94.31) for 20 of 30 days	From 20-Dec-2023 at 130% ($105.09)
Free conversion	From 1-Sep-2028	From 15-Sep-2025
Contingent conversion	130% price trigger, 98% trading-price trigger, corporate events	Same structure
Settlement	Cash up to principal; excess in cash or shares	Physical settlement eliminated by supplemental indenture (Nov-2023)

Term	2028 Notes	2025 Notes
Effective rate (ASU 2020-06)	1.1%	0.2%

Capped calls (2028 Notes). Uber paid about $141M for call spreads struck at $72.54 and capped at $95.81, a 75% premium to $54.75. They cover the 23.78M shares underlying the notes, and they offset dilution up to the cap. Uber recorded them in additional paid-in capital, with no remeasurement (10-K FY2024).

Fair value disclosures. Uber reports each note's fair value quarterly, as Level 2 "based on quoted prices in markets that are not active", rounded to $0.1bn: Q1, Q2 and Q3 2024 10-Qs, and the FY2024 10-K.

Accounting framework

The conversion option and the capped calls qualify for equity treatment. The notes are therefore a single liability at amortized cost, and fair value is a disclosure only.

Question	Analysis	Conclusion
ASC 480 scope	The notes are not shares and not an obligation to repurchase shares or deliver a fixed monetary amount in shares. The capped calls are assets.	Out of scope
ASC 815-15: embedded derivative?	The conversion option meets the derivative definition and is not clearly and closely related to the debt host.	Bifurcate unless a scope exception applies
ASC 815-40-15, step 1	The conversion contingencies (130% price trigger, 98% trading-price trigger, corporate events) refer only to Uber's own stock or the notes.	Pass

Question	Analysis	Conclusion
ASC 815-40-15, step 2	Settlement equals shares × price less fixed principal. Adjustments are standard anti-dilution and make-whole inputs to a fixed-for-fixed option.	Indexed to own stock
ASC 815-40-25	Uber controls share settlement of the excess. The share count is capped by the make-whole table, enough shares are authorized, and there is no net-cash settlement outside Uber's control.	Equity-classified
ASC 815-10-15-74(a)	Indexed and equity-classified.	Not bifurcated
ASC 470-20 (ASU 2020-06)	The cash-conversion and beneficial-conversion models no longer apply.	One liability; 1.1% effective rate
Capped calls	A freestanding purchased call spread on own stock, indexed and meeting the equity conditions.	APIC, not remeasured

What ASU 2020-06 changed. Under the legacy cash-conversion model, Uber would have split the 2028 Notes at issue:

- **Liability:** about $1,278M, the straight-debt value at the BB spread.
- **Equity:** about $447M.

The liability would then have accreted at about **7.1%** instead of 1.1%. The model computes this split; it is not a disclosed figure.

ASC 820. Fair value is the price to sell the note to a market participant at the measurement date. The notes trade over the counter, so Uber's Level 2 figures come from observed quotes. This model is the income-approach cross-check a valuation team would run on those quotes: an auditor's "independent estimate" of a Level 2 price.

Methodology

The notes are valued with a Tsiveriotis-Fernandes (1998) binomial lattice on Uber's share price. It splits each node's value into two parts:

- a **cash-only component** B (coupons, principal, call cash), discounted at the risk-free rate plus Uber's credit spread;
- an **equity component** V − B, discounted at the risk-free rate.

This prices credit risk on the bond-like part and none on the share-like part.

1. **Lattice.** A Cox-Ross-Rubinstein tree runs from the valuation date to maturity: 1,000 steps in Python and 100 in Excel, with an identical scheme. Coupons are paid at the first step on or after each coupon date.
2. **Terminal payoff.** At maturity, V = max(CR × S, $1,000) + coupon. B is $1,000 when the note redeems and zero when it converts.
3. **Backward induction.** The continuation value at each node is the r-discounted expectation of V − B plus the (r + s)-discounted expectation of B. Two rules then apply:
   a. **Soft call.** On or after the call date, if S ⩾ 130% of the conversion price, Uber calls and holders convert, so V = CR × S and B = 0.
   b. **Voluntary conversion.** Holders convert when CR × S exceeds the continuation value. With no dividend, this never happens before maturity.
4. **Inputs.** None is taken from the disclosed fair values:
   - **Spot:** Uber's month-end close.
   - **Risk-free rate:** the Treasury par curve, interpolated at the note's remaining term.
   - **Credit spread:** the ICE BofA BB option-adjusted spread before S&P's upgrade to BBB- on 16-Aug-2024, and the BBB spread after.
   - **Volatility:** 36 monthly log returns, annualized. This gives 45–46% at every 2024 date.

$V_k = e^{(-r\Delta t)} \cdot E[V_{k+1} - B_{k+1}] + e^{(-(r+s)\Delta t)} \cdot E[B_{k+1}]$, then $V_k = \max(CR \cdot S_k, V_k)$ and $B_k = 0$ wherever conversion or a forced call applies.

Checks.

- With the call and early conversion switched off, the lattice converges to the closed form $CR \cdot S \cdot N(d_1) + F \cdot e^{\{-(r+s)T\}} \cdot N(-d_2) + PV(\text{coupons})$, to within 0.05 per 100.
- The Excel lattice matches Python to four decimals on every case.

Alternatives considered and rejected.

- **Bond plus Black-Scholes call:** ignores the soft call (an error of 2–3% here) and applies one discount rate to both parts.
- **Single-rate (Goldman Sachs 1994) lattice:** blends credit and risk-free discounting by the conversion probability. TF is the market standard and separates the two cleanly.
- **Monte Carlo (Longstaff-Schwartz):** needed for path-dependent features such as averaging triggers. It is unnecessary for a one-factor note with a barrier-style call.
- **Two-factor (equity and credit) or jump-to-default models:** better for distressed issuers. Uber is BB+ to BBB, and the notes trade well above their bond floor.

Issue-date evidence (20-Nov-2023)

Two prices were set on pricing day: the capped calls and the notes themselves. Both are consistent with about 46% volatility once market conventions are allowed for.

Capped calls imply volatility skew, not a flat volatility. Uber paid $141M for 23.78M call spreads, or **$5.93 per share**. Under flat-volatility Black-Scholes, a 5-year $72.54/$95.81 spread on a $54.75 stock is worth at most **$4.90**, at 28% volatility. No single volatility reproduces the price paid.

Pricing the lower strike at the 46.3% realized volatility and the cap at 43.2% matches $5.93 exactly. That is **3.1 vol points of skew** across a 32% strike gap, which is normal for single-stock equity. The practical lesson for ASC 815-40 work on warrants and call spreads is that **flat-vol Black-Scholes understates call spreads** when skew exists.

The notes were sold below model value. At 46.3% volatility and the 2.51% BB spread, the lattice values the notes at about **103.6** per 100, against an issue price of 100 and net proceeds of 98.6. This new-issue discount is well documented in convertibles: it pays hedge funds to take the deal down and delta-hedge it.

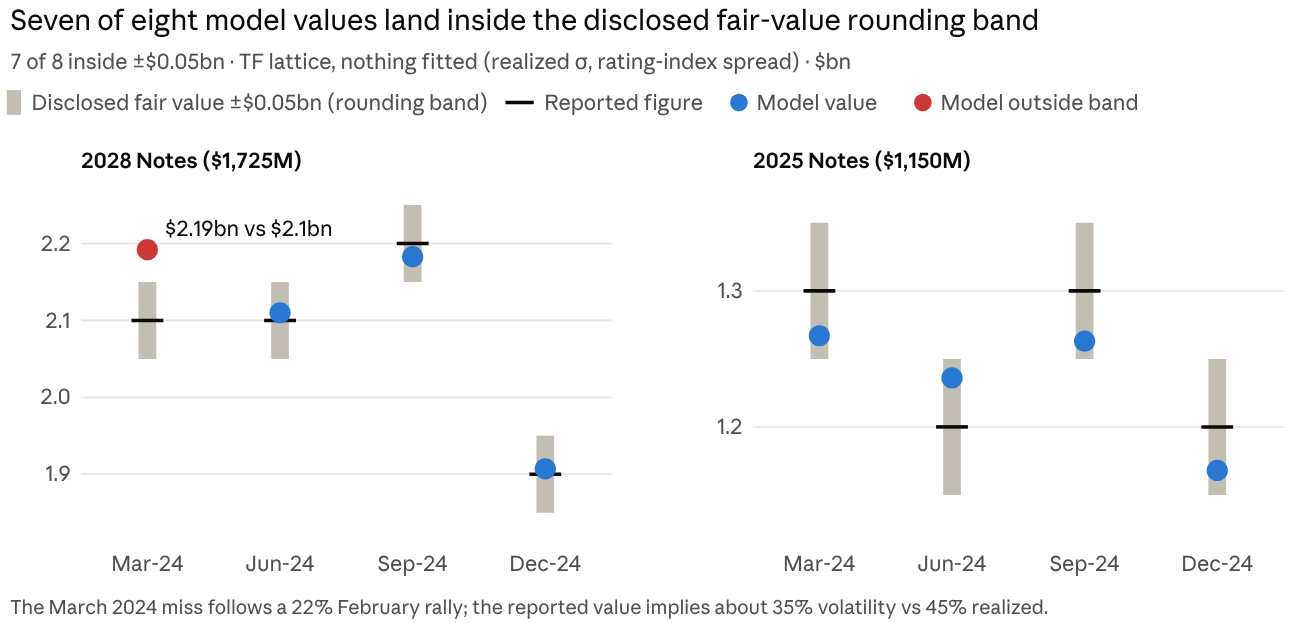
Seen the other way, par implies:

- **39%** volatility at the BB spread, or
- a **4.2%** credit spread at realized volatility.

Under ASC 820-10-30-3, day-one fair value equals the transaction price when nothing indicates otherwise. Every later quarter-end is tested against market-based disclosures (next section).

Quarter-end validation

The model reproduces seven of Uber's eight disclosed fair values for 2024, with a mean absolute error of 2.1% against the reported midpoints.



The March 2024 miss follows a 22% February rally; the reported value implies about 35% volatility vs 45% realized.

Model: convtf.py / uber_convertible_model.xlsx (1,000-step lattice). Reported: Uber 10-Q Q1–Q3 2024 and 10-K FY2024, Level 2, rounded to $0.1bn.

Why the test is demanding. Over the year, the 2028 Notes moved from 6% in the money (March) to 17% out of the money (December). The value split between bond floor and option changed a lot, and the model tracked it with fixed rules: realized volatility and rating-index spreads. The 2025 Notes sat near their bond floor throughout, so they mainly test the credit leg.

What the disclosures imply. Backsolving volatility from each reported figure (with the rounding band as a range) gives:

- **2028 Notes:** 35%, 45%, 47% and 44% (each about ±5 points), against 45–46% realized. A single volatility fitted to all four is about 43%.
- **2025 Notes:** consistent with realized volatility, but the $0.1bn rounding spans more than 30 vol points. These notes say little about volatility.

The March 2024 miss. The model gives $2.19bn against a reported $2.1bn (+4.4%). In February 2024 Uber reported its first full-year operating profit and its first share buyback, and the stock rose 22%. The reported value implies volatility of about 35%: the market priced calmer forward volatility than the trailing 36 months showed. A forward-looking (implied) volatility input, if available, would close most of the gap.

Sensitivities and value drivers

At 31-Dec-2024 the 2028 Notes were a **balanced** convertible. Their value of 110.8 per 100 splits into an 84.4 bond floor and a 26.5 conversion option. Delta is 0.47, so volatility and credit both matter.

2028 Notes fair value ($bn), 31-Dec-2024, by volatility and credit spread (reported: 1.9; band 1.85–1.95):

σ \ spread	50bp	75bp	102bp (base)	150bp	200bp
35%	1.839	1.831	1.823	1.808	1.793
40%	1.885	1.877	1.868	1.853	1.838
45% (base)	1.928	1.919	**1.911**	1.895	1.879
50%	1.969	1.960	1.952	1.937	1.921

- **Volatility:** ±5 vol points move the value by about ±$0.04bn, close to a full rounding band. Vega is 0.16 per 100 per vol point.
- **Credit:** 50bp moves the value by about $0.02bn (CS01 ≈ −0.018 per 100 per bp). Credit matters less than for straight debt because half the value is equity-like and discounted risk-free.
- **Uber-specific credit:** Uber's 4.30% 2030 senior notes priced at **T+75bp** on 4-Sep-2024 (term sheet), 61% of the BBB index that day. Scaling the post-upgrade spreads by that ratio raises the September and December values by 0.6–0.7%. All the conclusions hold.
- **Soft call:** ignoring it overstates the 2028 Notes by 2–3% in every quarter. This is the main error of a "bond plus Black-Scholes call" shortcut.
- **Lattice steps:** 100 steps (Excel) against 1,000 differ by 0.0–0.8 per 100, largest in March 2024 when the stock was near the conversion price.

If the option had failed equity classification. A bifurcated conversion option would have been worth $840M, $740M, $710M and $450M at the four 2024 quarter-ends (100-step lattice). The quarterly swings of $30–260M would have gone through earnings. This is why the ASC 815-40 analysis is the first question on any convertible engagement.

Mock auditor review: comments and responses

#	Auditor comment	Response
1	Why realized rather than implied volatility?	Implied volatility for 3–5 year Uber options was not available from public sources. Realized volatility keeps the test fully out-of-sample: no input is taken from the figures being tested. The volatility backsolved from the 2028 Notes (35–47%) brackets the realized 45–46%. In practice, Bloomberg OVDV long-dated implied volatility would be the primary input.
2	Monthly returns give few observations.	36 observations give a standard error of about ±5 vol points. The 24-month window gives 41–49%, and the capped-call price independently supports 46% with skew.
3	Why index spreads instead of Uber's own curve?	Uber's only directly observable spread is the T+75bp new-issue print in September 2024. The rating index gives a consistent series across the upgrade. Using the Uber-specific ratio changes values by under 1%.
4	The soft call has a 20-of-30-day averaging condition and a make-whole on redemption.	The lattice treats 130% as a hard barrier. This slightly understates value near the trigger, but the stock stayed below $94.31 throughout 2024 and the call window opens only in December 2026.
5	Contingent conversion is not modelled.	With no dividend, early conversion is never optimal, so allowing conversion at every node changes nothing. The contingency affects balance-sheet classification (current vs long-term), not value.

#	Auditor comment	Response
6	The March 2024 2028-note value is 4.4% above the disclosure.	Accepted as a model limitation. The reported figure implies 35% volatility after a 22% rally. A valuation would document the volatility choice against implied data at that date.
7	Issue-date model value of 103.6 exceeds the transaction price.	ASC 820-10-30-3: transaction price is the day-one fair value. The 3.6-point gap is the documented new-issue discount and is not a day-one gain.
8	Support equity classification of the capped calls.	Freestanding, indexed to Uber stock (fixed strike and cap, standard adjustments), and settled in shares or net cash at Uber's election. Recorded in APIC and not remeasured, consistent with Uber's disclosure.
9	Excel and Python could diverge.	The Results sheet reconciles every case to four decimals (difference 0.0000). The notebook reports the 100- vs 1,000-step discretization.

Open items: obtain long-dated implied volatility and Uber CDS or bond quotes for each date, and confirm the make-whole table against the indenture.

Limitations and sources

Limitations

- **Rounded disclosures.** Uber reports fair values to $0.1bn, so the test is against a band. The 2025 Notes constrain volatility only loosely.
- **Volatility input.** Volatility comes from monthly closes, not an implied surface. The issue-date rate is a monthly average (FRED GS5, Nov-2023) because the daily curve for 20-Nov-2023 was not retrievable.
- **Simplified features.** The soft call is a hard barrier, with no averaging and no make-whole. Fundamental-change puts, contingent conversion and borrow cost are not modelled. Credit uses rating indices, not Uber CDS.

- **Purpose.** This is an independent re-performance from public information for educational purposes. It is not a valuation opinion and is not affiliated with Uber or its auditors.

Sources

- Uber pricing release, 2028 Notes (20-Nov-2023) and 8-K
- Uber pricing release, 2025 Notes (8-Dec-2020)
- Uber 10-Qs: Q1 2024, Q2 2024, Q3 2024; 10-K FY2024
- Uber FWP: 2030/2034/2054 senior notes (4-Sep-2024)
- S&P upgrade to BBB- (Aug-2024); Uber's inaugural IG rating (Fitch, Moody's)
- US Treasury daily par yield curve, 2024; FRED GS5
- FRED ICE BofA BB OAS; FRED ICE BofA BBB OAS
- Uber month-end prices (digrin)
- Tsiveriotis, K. and Fernandes, C. (1998). Valuing convertible bonds with credit risk. *Journal of Fixed Income*, 8(2), 95–102.